PROFESSIONAL EDUCATION




SUMMER 2024

# ADVANCED DATA ANALYTICS FOR IIOT AND SMART MANUFACTURING

| DATES | DURATION | FORMAT | FEE | CEUS |
|---|---|---|---|---|
| June 24–27, 2024 | 4 Days | On Campus | $3,900 | 2.5 |

## Explore the latest developments in smart manufacturing

This course is designed to help you learn and apply advanced data tools for IIoT and smart manufacturing. The curriculum ranges from foundational concepts to in-depth, hands-on activities using production data, and covers a variety of cutting-edge approaches such as deep reinforcement learning control, encryption for data outsourcing, and predictive data analytics algorithms.

Through interactive lectures and exercises—grounded in data and examples from the automotive, semiconductor, food production, nanofabrication, and telecommunications sectors—you will acquire core strategies and frameworks for using data driven analysis, simulation, automation, and optimization techniques to improve manufacturing processes and deploy IIoT systems. You will then put your learning into action by working with other course participants to design, simulate, and test system improvements based on information extracted from process and production data.

The curriculum is organized into four sections:

- Data collection strategies for IIoT
- Data analytics and predictive modeling for production data
- Simulation modeling through digital twinning
- Process and manufacturing systems simulation and optimization

### WHO SHOULD ATTEND

- **Engineers and engineering managers** who want to implement the latest digital strategies to maximize the value of their operations
- **Design and manufacturing engineers or machine and process designers** seeking to enhance their ability to capitalize on data and modeling in a manufacturing environment
- **Data scientists** looking to take the lead in the rapidly growing field of smart manufacturing
- **Factory, production, or process managers** who want to use advanced tools to design and deploy integrated, computational systems on the factory floor
- **Directors of operations or manufacturing** who would benefit from implementing data analytics tools as part of a continuous improvement strategy
- **Consultants** who are interested in increasing their value to clients in the manufacturing industry

### EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Design & Manufacturing***.

*professional.mit.edu/aism*

## LEARNING OUTCOMES

- Develop a data collection strategy for industrial networks, including non-traditional IoT data streams and cloud-based data storage
- Visualize, analyze, and predict relevant production outcomes based on process data
- Create and use digital twins of factory processes
- Simulate and optimize factory processes using data analytics, predictive modeling, and digital twins

## DISCUSSION AND CASE STUDY TOPICS

You will leave this course equipped to capitalize on the latest data-driven manufacturing tools, including video analytics, learned control systems, and system security strategies.

- **Data-driven manufacturing planning and execution analytics and services:** Providing new approaches and models for manufacturing decision making in smart factories.
- **Data-driven manufacturing asset analytics and services:** Providing new whole-life asset management models and tools in smart factories.
- **Data-driven manufacturing quality analytics**
- **Data-driven smart manufacturing analytics dashboard:** Providing visibility methods and tools for smart manufacturing analytics platform.

## LEAD INSTRUCTOR

### Brian Anthony



Dr. Anthony is Director of MIT's Master of Engineering in Manufacturing Program, Co-Director of the Medical Electronic Device Realization Center, and Associate Director, MIT.nano. With over 20 years experience in product realization—Dr. Anthony won an Emmy (from the Academy of Television Arts and Sciences) in broadcast technical innovation—Dr. Anthony designs instruments and techniques to monitor and control physical systems. His work involves systems analysis and design and calling upon mechanical, electrical, and optical engineering, along with computer science and optimization, to create solutions.

The focus of Dr. Anthony's research is in computational instrumentation—the design of instruments and techniques to measure and control complex physical systems. His research includes the development of instrumentation and measurement solutions for manufacturing systems and medical diagnostics and imaging systems.

## Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

PROFESSIONAL EDUCATION

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